

Machine Learning and its Applications

Today's goals

Module 1

Machine Learning Fundamentals I: Basic Concepts

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 8 Introduction

Introduction to Module 8's Study

Video • 1 min

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What Does “Good” Mean in Machine Learning Models?

Reading • 5 min

Module 8-1 ML Model Training Fundamentals

ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions

Video • 8 min

ML Model Training Fundamentals: Loss Functions and Gradient Descent

Video • 5 min

ML Model Training Fundamentals: Gradient Descent

Video • 10 min

Materials for ML model training

Reading • 10 min

What Does It Actually Mean to “Train” a Machine Learning Model?

Reading • 5 min

Module 8-2 ML Model Evaluation Fundamentals

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 13 min

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 5 min

Materials on ML Model Evaluation Fundamentals

Reading • 10 min

When a Model Fails: Is It Bias, Variance, or Something Else?

Reading • 5 min

Module 8-3 Summary of the ML Process

Summary of the ML Process

Video • 5 min

Materials for ML Process

Reading • 10 min

From Data to Decisions: Reconstructing the Machine Learning Pipeline

Reading • 5 min

Module 8 Learning Coach Dialogue

• 5 min

Module 8 Quiz

Module 8 Quiz

Practice Assignment • 30 min

Module 8 Assignments

Module 8 Assignment 1

Graded App Item • 30 min

Module 8 Assignment 2

Graded App Item • 30 min

CSV file for Module 8 Assignment

Reading • 1 min

Module 8 Recap

Reading • 5 min

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

What Does “Good” Mean in Machine Learning Models?

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In the previous module, you learned how to represent data and build machine learning models. This module shifts focus to a more difficult question: **how do we judge whether a trained model is actually good?**

Reflect on the following:

1. Model strength vs model quality

You are using a powerful machine learning method and obtain a trained model.

- Is high model quality guaranteed simply because the method is strong?
- What factors *outside the algorithm itself* can cause a powerful model to perform poorly or give misleading results?

2. Training vs real-world performance

A model may perform very well during training.

- Why might this performance fail to generalize to new, unseen data?
- What risks arise if model quality is judged only using training results?

3. What should be evaluated—and why

As engineers or scientists, trust matters.

- What aspects of a model's behavior would you want to evaluate before trusting its predictions?
- How might different applications (e.g. finance, healthcare, engineering design) change what “good quality” means?

4. Looking ahead

This module introduces concepts such as **loss functions, optimization, overfitting, underfitting, bias, and variance**.

- Which of these ideas do you expect to be most critical for assessing model quality, and why?
- What questions do you currently have about evaluating machine learning models reliably?

Go to next item

✔ Completed

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https://www.coursera.org/learn/machine-learning-and-its-engineering-applications/supplement/WqpLP/what-does-good-mean-in-machine-learning-models

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